COMP396 Weekly Meeting Minutes

(Week 3)

Date & Time: Sunday, October 11, 2025, 17:30-20:30

Location: Sydney Jones Library / Abercromby Study Room 1

Note Taker: Wong Haw

Attendees: Wong Haw, Zhang Olivia, Xue Yudan, Zhao Saria

1. Agenda:

- Check the completion of the Phase 2: Core Feature Implementation task (B1-B4).
- Share our learning experience and any idea that would be benefits to the group project from the Backtrader framework and core teaching materials (Lectures 05 & 06).
- Each member presents and discusses their personal strategic ideas (B4).
- Discuss specific methods and technical paths to determine trading strategies.
- Plan the next phase of tasks and assign them specifically.

2. Discussion & Decisions:

2.1. Task Review and Result Confirmation

Our team confirmed that all core tasks (B1-B4) assigned at last week's meeting were completed on time.

- **Core Knowledge Enhancement (B1):** All members have completed self-study of Lecture 03 and deepened their understanding of budget constraints, bankruptcy rules, and order types.
- **Core Function Implementation (B2):** All members completed the relevant exercises, expanded their data processing capabilities to the OHLCLV format, learned how to integrate `technical indicators using pandas-ta`, and applied `optstrategy` for parameter optimization.
- **Framework Review (B3):** All members systematically reviewed Backtrader's official documentation.
- **Preliminary Strategy Ideation (B4):** All members prepared a preliminary idea of their personal strategy and shared it in this meeting.
- **Environment Setup:** Our team additionally completed all the environment setup tasks for the "Backtester Guide" on Canvas, paving the way for subsequent policy development and testing.

2.2. Technical Learning and Progress

Our team focused on the recently uploaded teaching materials, focusing on the following:

- **Lecture 05: Position Sizing:** All team members learned the importance of position sizing in depth. It is clarified that the final performance is closely related to the contribution of different series, and several methods for determining sizing are studied, such as equal position, inversely proportional to the opening price, and inversely proportional to the average absolute difference. The team recognised that a good strategy needed to combine a suitable position sizing approach with money management (e.g., scaling up capital to £900,000).
- **Lecture 06: Graphics:** Our team studied Lecture 06 together, reviewing `matplotlib` and `seaborn` application in financial data analysis. The focus is on how charts can be used to

assist in the development and presentation of strategies, such as by plotting overlays of prices and indicators such as Bollinger Bands, equity curves and underwater curves, and parameter-optimized heat maps (Fitness Landscapes) to visually evaluate strategy performance.

2.3. Strategy Selection Methodology Discussion

Based on the initial idea, our team had an in-depth discussion on how to choose the most appropriate strategy for a specific data series and came up with two main ideas:

- **Empirical observation method:** Allocate the same initial virtual funds to multiple alternative strategies, backtest on all data series, and judge the match between the strategy and the data series empirically by observing their performance indicators such as PnL curve and PD ratio.
- **Data-driven approach:** Prioritize writing code to analyze all data to identify the type of market (e.g., trending or volatile). It then automatically matches theoretically better strategies based on market type. For example, the team discussed the application of trend-following when a clear upward or downward trend is identified, and the application of mean-reversion in a consolidation-volatile market.

We agreed that data-driven methods are more systematic and scalable, but they also pose greater technical challenges. It was decided to explore at the same time in the next stage.

3. Task Allocation:

To efficiently advance the preparation of the first report, our team has developed a task plan for the next week:

ID	Module	Task Description	Owner	Due Date
C1	Strategy	According to the structure requirements of the final report, each member completes the	All	Oct. 17

	Design	initial conception and code writing of three independent strategies, including the strategy principle, entry and exit conditions, and applicable markets.	Members	
C2	Data Analysis & Visualization	In-depth practice of all the chart in Lecture 6 making methodology: 1. Understand the chart composition: Reproduce each chart code in L6 to understand its data source and plotting logic. 2. Interpret chart information: In-depth analysis of the function of each chart (such as price and earnings charts, equity and underwater curve charts, correlation heat maps, parameter-optimized heat maps, etc.), understand what market characteristics or strategy performance can these charts reveal and provide decision-making support for strategy selection.	All Members	Oct. 17
C3	Project Management	Organize specific questions your team encounters in strategy design and technical implementation and schedule a meeting with the Module Leader for a focused Q&A session.	Yudan Xue	Oct. 17

4. Challenges and Issues

- **Strategy Choices:** We feel overwhelmed by the vast array of indicators and different strategy types. The main difficulty lies in the inability to clearly define the three different

market environments (e.g., trends, oscillators, etc.) and which trading strategies should be prioritized and tested in these environments, resulting in a lack of a clear starting point in the strategy ideation phase.

- **Diagram application confusion:** Lecture 6 introduced a large number of visual charts, which while powerful, also made the team feel confused. The team is currently unable to determine which charts are most critical and necessary for identifying market characteristics and screening initial strategies at this stage, fearing that too much time will be spent on generating insufficient charts.

5. Supervisor Question

- The project progressed smoothly, but the team formed some specific problems in the details of the strategy design and the technical implementation path. After the plan is sorted out next week, take the initiative to communicate with the supervisor.

6. Next Meeting Arrangement

- **Date & Time:** Friday, October 17th, 2025, 14:00-17:00
- **Location:** Sydney Jones Library / Abercromby Study Room 1
- **Key Agenda Items:**
  - Review and discuss the preliminary strategy conception documents (C1) from each member.
  - Share insights and discoveries on data analysis using advanced charts (C2).
  - Finalize the list of questions that need to be consulted with the Module Leader
  - Plan the next phase of the task: initial coding implementation and backtesting of the policy.